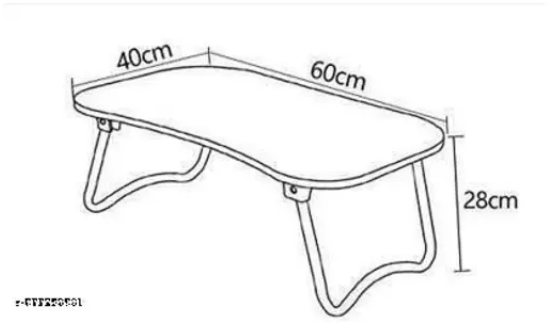


Sahrdaya College of Engineering and Technology, Kodakara	
Answer Key	
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY	
SECOND SEMESTER B.TECH DEGREE EXAMINATION, JUNE 2023	
Course Code: EST200	
Course Name: Design and Engineering	
PART A Answer all questions, each carries 4 marks	
Q1	How is engineering design different from other kinds of design?
ans	An engineer is a creative, ingenious person and Engineers create ingenious solutions to societal problems. Engineering design process can be defined in many ways as follows. Engineering design is a systematic, intelligent process in which engineers generate, evaluate, and specify solutions for devices, systems, or processes whose form(s) and function(s) achieve clients' objectives and users' needs while satisfying a specified set of constraints. In other words, engineering design is a thoughtful process for generating plans or schemes for devices, systems, or processes that attain given objectives while adhering to specified constraints. The engineering design process is a methodical series of steps that engineers use in creating functional products and processes. It is a decision-making process, in which the basic sciences, mathematics, and engineering sciences are applied to optimally convert resources to meet a stated objective. Among the fundamental elements of the design process is the establishment of objectives and criteria, synthesis, analysis, construction, testing, and evaluation.
Q2	Explain how to identify customer requirements of design.
	1.12.1 Need identification The design process starts with need identification. At times the user, the client and the customer could be the same or different. To start with, the Designer has to meet the needs of the Client as well as the User. The entity that has the need to use the design is the user. For any Design to be taken up, there should be a need gap or a problem that needs a solution. This gap or need could be identified by the user, an observer or by an organization. Through interviews with potential purchasers, focus groups, and by observing similar products in use, researchers identify these gaps or user needs. The need for a new design can be generated from several sources, including the following: 1. Client request – In a design company, a client may submit a request for developing an artifact. It is often unlikely that the need will be expressed clearly. The client may know only the type of product that he or she wants; for example, “I need a safe ladder.”

	2. Modification of an existing design – Often a client asks for a modification of an existing artifact to make it simpler and easy to use. In addition, companies may want to provide customers with new, easy-to-use products. 3. Generation of a new product – In all profit-oriented industries, the attention, talent, and abilities of management, engineering, production, inspection, advertising, marketing, sales, and servicing are focused on causing the product to return profit for the company and in turn for company stockholders. For an industry to survive in today's world, it must continue to grow; it cannot afford to remain static. This growth, throughout history, has been built on new products. A product will peak out when it has saturated the market and then begin to decline. It is obvious that an industry must seek out and promote a flow of new product ideas. These new products are usually protected by applying for patents. The needs of customers outside of the company are important to the development of the product design specifications for new or improved products. A second set of critical constituents are the internal customers, such as a company's own corporate management, manufacturing personnel, the sales staff, and field service personnel must be considered. For example, the design engineer who receives information on the properties of three potential materials for his or her design is an internal customer of the materials specialist.
Q3	Explain the term design functions as applicable to engineering design.
	2.3.3 Design functions Functionality is the ability of a design to meet its product's operational criteria. The product designed must do its intended job. If the product is a chair, it has to hold your backside off the ground. If it's a table, you must be able to sit at it, and you must be able to lie in a bed. Some designs may have several functions. The main function termed the primary function and the other functions the secondary functions which may include safety, strength, ease of use, efficiency, simplicity, durability etc. Commercial product design often has a strong emphasis on functional efficiency that places aesthetics behind the qualities of easy, reliable operation. However, in the majority of consumer products, aesthetic qualities and image drive the market, often at the expense of function.
Q4	How does design thinking approach help engineers in creating innovative and efficient designs?
	Design thinking is a process for solving problems by prioritizing the consumer's needs above all else. It relies on observing, with empathy, how people interact with their environments, and employs an iterative, hands-on approach to creating innovative solutions. Design thinking is "human-centered," which means that it uses evidence of how consumers (humans) actually engage with a product or service, rather than how someone else or an organization thinks they will engage with it. To be truly human-centered, designers watch how people use a product or service and continue to refine the product or service in order to improve the consumer's experience. This is the "iterative" part of design thinking. It favors moving quickly to get prototypes out to test, rather than endless research or rumination. In contrast to traditional problem-solving, which is a linear process of identifying a problem and then brainstorming solutions, design thinking only works if it is iterative. It is less of a means to get to a

	single solution, and more of a way to continuously evolve your thinking and respond to consumer needs.
	PART B
	Design a study table for a child upto 5 years old specifying objectives, functions and constraints through various stages of design processes. Use hand sketches to illustrate the process.
	Problem/need To design a chair for a 5-year-old child Objectives • chair must be safe for children • easily accessible for 5 year old child • god material must be used (child safe) • must be strong • steady structure • should be able to have food while sitting Constraints (the limitations to the design) • Height of the chair must be between 30 to 40 cm • Budget must be minimum (below 500) • Material used must be strong and light weight • Safety must be priority Design 1  ✓ Material used – Plastic ✓ Height of the table – 28 cm ✓ foldable legs are available ✓ Detachable ✓ Budget is nearly Rs 400

✓ Colours available – blue, red, yellow, green

Highlight of the design

The design 1 satisfies most of the objectives. The material used is safe for kids and its light weight. The plastic used material is safe for kids. The height of the chair is between the suggested height mentioned in the objective. There is a detachable food table with this chair, with no extra cost. So it can be used as a study table also. A variety of colours are available with this design.

Design 2



✓ Material used – wood

✓ Height of the chair – 40 cm

✓ Two closed shelves are available

✓ Budget is – 700 rs

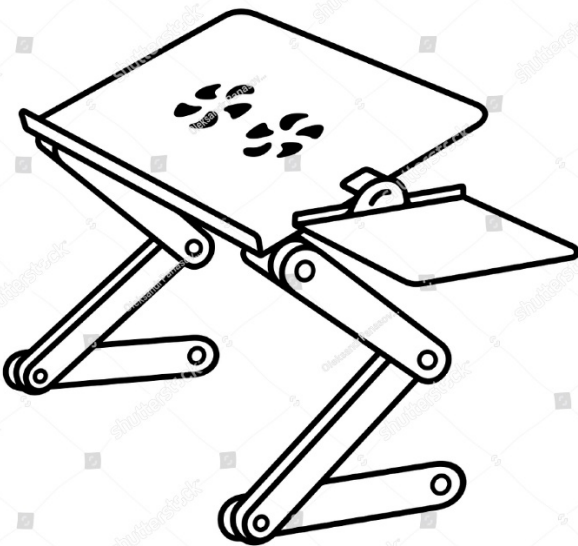
✓ Colour available – wood colour only

Highlight of the design

In this design the material used is wood. The height of the chair is 40 cm. the wood material colour only available in this design. The chair base is available with cushions for comfort

In the above two designs each design meets most of the objectives/ requirements. By comparing these two designs design 1 have good plastic material used and its light weighted. Which is easy for children's use. Also its available in different colours. The design 2 is on the other hand little bit more expensive and not a good design for children. The weight of the table is more, compared with the design 1. Also the food table need to be purchased additionally. The design 1 meets more merits to the given objectives, so we can finally choose design 1 as our preferred design. Design 1

Choose the best design for a laptop stand for avoiding overheating, also

	incorporate the customer requirements. Show how design objectives were finalized considering the constraints also.
	Problem/need To design a laptop stand for avoiding overheating Objectives • avoiding overheating • easy to use • good material must be used • must be strong • steady structure Constraints (the limitations to the design) • avoiding overheating • Budget must be minimum (below 500) • Material used must be strong and light weight • Safety must be priority Design 1  shutterstock NAME ID: 156417193 www.shutterstock.com Colour Silver Form Factor Aluminum alloy Compatible Devices Laptops, Tablet, Books, Projector, MacBook Pro/Air, Dell XPS, HP, Samsung Laptops and Other 11-17 Inch NotebookLaptops,

	Tablet, Books, Projector, MacBook Pro/Air, Dell XPS, HP, Samsung Laptops and Other 11-17 Inch Notebook Mounting Type Tabletop Highlight of the design Ergonomic Design - The adjustable stand elevates your laptop by 10.6 inches to a perfect eye level and prevents you from hunching over your screen, which reduces neck and shoulder pain. Moreover, the edge of the stand is so smooth that protects you from being scratched. Sturdy and Foldable - The office computer stand is made of thickened aluminum alloy, so the stand is sturdier and less wobbly. The Non-slip Silicone pads stick tightly and ensure your laptop stable on the stand. And the rubber feet prevent the stand from slipping on your desktop. And the lightweight laptop stand is Foldable. Wide Compatibility - This laptop riser is compatible with all laptop models and sizes up to 17 inches, such as MacBook Pro/Air, Surface Laptop, Dell XPS, HP, Samsung, Lenovo, Google Pixelbook, Acer, Chromebook and more. Ventilation Design - EPN laptop riser is made of aluminum alloy material, sturdy, rust-proof; scratch-proof, the heat conduction alloy and ventilation hollow design, work together to cool down your computer and prevent overheating. Multi-Use & Practical - Multi-use on many occasions such as office, meeting room, library, coffee shop, kitchen and music playing house to hold the laptops, tablets, projectors, menu, music score, magazines etc. An excellent useful gift for your friends and family. Design 2  Colour Dark Grey
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		Form Factor Table Stand Compatible Devices Laptops Mounting Type Tabletop Highlight of the design Made of Aluminum Alloy with Anti Slip Silicon Rubber pads to keep laptop in place Seven Adjustable Levels It Supports Upto 5kg Weight Premium and Stylish Design Foldable Stand for Laptops up to 17 inches (Including Dell, HP, Lenovo, Macbook, Mi, Surface, Thinkpad, Asus, etc) Portable and Easy to Carry with the Pouch Open heat dissipation - helps in Preventing Overheating In the above two designs each design meets most of the objectives/ requirements. By comparing these two designs design 1 is adopted
		Construct a number of possible designs and then refine them to narrow down to the best design for a baby seater in a car. Show how convergent-divergent thinking helps in the process. Illustrate with hand sketches.
		Design a carry bag for shopping purposes. Illustrate the various stages involved in design thinking. Sketch the final design.